

Student Name:

Weight: 35%

Student ID:

Marks: /32

Deliverable 4: Analysis

Introduction

This capstone project for the Data Analytics program gives you an opportunity to synthesize all aspects of your previous courses in a final, complete data analysis solution. You'll use all your knowledge to build an end-to-end solution and present it to a "client" (your classmates, instructor and possibly industry guests). You'll use data analysis to provide real value to an organization or business.

Phase 4: Building the Analytical Solution

In this phase, you'll build your solution and perform your analysis of the data using the tools of your choice (which should have been documented in your proposal). This analysis should be comprehensive and meet the organizational needs you defined in your proposal.

Instructions

1. Build the solution that you defined in your proposal using the tools you chose. When completing your analysis, ensure that:
 - a. Your documentation (**15 page maximum including figures**) is clear and easy to understand. Your documentation must include:
 - i. An overview of the project including a description the problem that you are attempting to solve
 - ii. A description of the solution you built including:
 1. A data dictionary with the fields, data types, and descriptions of the data you used and a diagram of the data model showing any relationships used
 2. A description of the product you created (for example, a Power BI report, Python notebook, Excel spreadsheet, etc.) and how it works and sample of the output it produces (for example, a visualization or a prediction based on the type of solution it is)
 - iii. A summary of the results of your analysis describing how your solution meets the objectives of your proposal
 - iv. Any changes to the original proposal that you made to complete the project including changes to the project scope and analysis method
 - b. Sources are properly cited and referenced throughout

- c. The risk/reward is managed
 - d. The solution meets the objectives of the capstone plan/proposal to serve the organization/business needs
 - i. There must be a clear relationship between your solution and the problem you initially set out to solve
 - ii. The solution must be appropriate for the type of analysis you set out to perform
 - e. All elements of the solution convey accurate and meaningful information in an appropriate format:
 - i. Any visual elements or models produced must be chosen appropriately for the data used and convey useful information relevant to the goals of the project
 - f. There is evidence of critical thinking and analysis
2. Submit all files and documents related to your solution to the Brightspace dropbox including:
- a. Your solution documentation (as a Word Document or PDF)
 - b. All components of your solution which may include:
 - i. Python/R/SQL scripts and notebooks
 - ii. Excel spreadsheets
 - iii. Power BI reports
 - iv. Copies of your raw data or links to the data sources

Marking Criteria

Criteria	Marks	Total Marks	
There is evidence of critical thinking and analysis	5	/5	
Documentation is complete, clear, and easy to understand	8	/8	
The solution meets the objectives of the capstone plan/proposal to serve the organization/business needs	12	/12	
Visual elements/models are appropriate and convey information to the organization/business	5	/5	
Grammar and writing are clear	2	/2	
Marks: 0 – Below Average 18 – Average 23 – Good 28 – Exceptional		Total	/32

Comments